



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli uczenia głębokiego między symulacją a rzeczywistością w fizycznej sztucznej inteligencji / Representation learning methods for simulation-to-reality generalization of deep learning models in physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	Doctoral School courses. Reading the literature and considering possible research directions
Semestr 2	Doctoral School courses and literature review. Choosing the dissertation topic with the supervisor and preparing the Individual Research Plan
Semestr 3	Preparing the research environment with digital twins, simulators and public datasets. Initial experiments on harmful reconstruction errors (RQ1)
Semestr 4	Further experiments on RQ1. Preparing and submitting the first publication to a 200-point venue in machine learning or robot learning, such as ICLR, RSS or IEEE RA-L. First experiments on RQ2
Semestr 5	Experiments on localizing the gap inside models and aligning representations (RQ2). Preparing a scientific publication
Semestr 6	Experiments on selecting real data (RQ3) and on transfer to unseen scenes and between tasks (RQ4). Preparing a publication and, if access allows, tests on a real robot
Semestr 7	Final experiments and submitting a paper to a conference or journal. First draft of the dissertation
Semestr 8	Completing and revising the doctoral dissertation, then submitting it

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission



Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Robots that learn by trial and error need a very large number of attempts, more than is practical or safe to collect on a real robot [12]. For this reason such models are usually trained in simulation [17]. A newer kind of simulation is the digital twin. A short video of a real room or table is turned into a photorealistic 3D copy with methods such as 3D Gaussian Splatting. The robot model is trained in this copy and then used in the real place.

The weak point of this approach is that the copy is never exact. A glass door may be missing from the copy, a mirror may look like an open room behind it, and textures and lighting are always slightly different. A model that works well in the copy can therefore fail in reality [16], and it does even worse in real places for which no copy was made [17]. This drop in performance is called the simulation-to-reality gap, and it is one of the main reasons why learned robot models are still hard to deploy.

The usual remedies do not use knowledge of where the copy is wrong. Domain randomization adds random changes to colours, textures or physics [5]. Newer variants learn how wide these changes should be [6],[7], but they tune physical parameters and do not ask which visual errors of the copy actually matter. Other methods force the model to see simulated and real images the same way, but in doing so they can also remove information the robot needs. When a few real examples are collected, they are mostly used to correct either the copy [6],[23] or the model [9],[1], and I found no method that chooses them to improve both at once. Earlier studies measured how factors such as lighting, texture or physics realism affect transfer [19],[25], but it is still not known which local reconstruction errors of a digital twin harm a model and at which stage inside it the harm appears.

I want to answer this question by looking inside the models. A neural network does not use the image directly. It turns it into internal features, called representations, and makes its decision from them. My working assumption is that an error in the copy that leaves these features unchanged does little harm, while an error that changes the features the decision depends on is likely to cause failures [3],[21]. This gives a measurable way to study the gap. First, I will check which reconstruction errors change the representations in a harmful way. Then I will find the stage inside the model at which real images lose information needed for the task, and correct the representations there. I will also study how to choose the few real examples that best improve both the copy and the model, and whether the improvements hold in new places and in a second task. I will work on two tasks of equal weight, robot navigation and robotic manipulation, and always test on scenes not used for training.

The answer would help decide which parts of a digital twin need to be accurate and where extra real data is worth collecting. In practice, a robot could be trained in a digital copy of a new warehouse, hospital or home and work there reliably after little real-world adaptation. The same questions arise



in autonomous driving, where simulators and reconstructions of recorded drives are widely used to train and test models.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Deep learning models for robot navigation and manipulation need more experience than real robots can easily provide, so they are often trained fully or partly in simulation [1]. A study on navigation showed that performance in simulation can poorly predict performance in reality unless the simulator is carefully tuned [2]. The simulation-to-reality gap is an instance of distribution shift. Domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions and the joint error of the best single model on both domains [3]. I treat this bound only as motivation. It suggests making the training distribution cover reality and making the model insensitive to the remaining discrepancy. Invariant features do not target the joint error, however, and can even increase it [4]. Correcting the simulation towards reality changes the training data itself and may therefore reduce this error, because it removes views whose correct action differs between the twin and reality.

A widely used approach is domain randomization, which varies simulator parameters such as textures so that reality appears to the model as one more variation [5]. Randomizing too widely leads to conservative behaviour, so later works shape the distribution by fitting it to a few real rollouts [6] or by maximizing its entropy while preserving task success [7]. These methods sample or adapt global parameters of a hand-built simulator, such as textures, lighting, friction or masses [5],[6],[7]. They do not use knowledge of where a scene is reconstructed wrongly. Twins built from real data have errors that vary strongly from region to region, for example where few camera views cover the scene, and global parameters cannot describe such errors.

Another approach makes the model invariant to the domain, so that simulated and real inputs give similar features. Domain-adversarial training makes the features of the two domains indistinguishable to a domain classifier [8]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved the real-world performance of co-trained policies, that is, models that map observations to actions [9]. However, invariance with low source error does not guarantee transfer. It can increase the joint error term when the label distributions, here the actions, differ between domains [4], and full invariance may discard task information. A recent analysis showed that effective co-training on simulated and real data aligns the two domains but keeps them distinguishable [10]. Whether the domains can be told apart is therefore not a measure of harm, and a useful measure must refer to the information the task needs. In sim-to-real work, invariance is usually enforced at a place chosen in advance, such as the input images, the final features [8] or a shared latent space [9], without first measuring where the gap arises.

Digital twins built from real data are a more recent direction. 3D Gaussian Splatting [11] reconstructs photorealistic scenes from multi-view photographs and renders them in real time. In manipulation, twins have been used to make policies learned from a few real demonstrations more robust through reinforcement learning in a quick scan of the target scene [12], to transfer image-based policies to reality without real-world demonstrations [13] and to guide real-world reinforcement learning from a phone capture [14]. The SIMPLER study [15] evaluated the same policies in simulation and on two real robot setups. In scenes visually matched to real images and with matched controllers, simulated success rates correlated strongly with real ones and ranked the policies in nearly the same order.



PolaRiS [16] turns short video scans of real scenes into interactive simulation environments for evaluating policies, but it still needs to co-train the policies on simulated data to bridge the gaps that remain after reconstruction, so even careful twins keep errors that matter for policies. In navigation, policies have been fine-tuned in twins built from a phone capture of a room [17] and trained in twins built from monocular urban videos [18]. Simulation factors such as randomization, photorealism and physics realism have mostly been varied as one setting for the whole scene, as in a recent study of real manipulation [19]. For whole workspaces, the similarity of pretrained features between rendered and real views was more strongly associated with the agreement of simulated and real policy performance than image quality metrics such as LPIPS or PSNR [20]. It is still open which local reconstruction errors harm generalization in navigation and manipulation, and how strongly each region should be randomized given its reconstruction error and its relevance to the task.

It is also unclear where inside a model the gap arises. Probing, which reads out information from hidden states with simple classifiers, showed that fine-tuning for actions degrades the visual representations of vision-language-action models [21]. Another study fine-tuned only selected layers and found that the best layers to adapt depend on the type of shift [22]. A study of co-training compared simulated and real features across layers, but it measured how well the two domains align rather than whether task information survives [10]. I found no work that locates, in models trained in twins, the stage at which task information available in simulation stops being recoverable from real inputs.

The remaining gap is usually reduced with a small amount of real data. In manipulation, co-training with simulated data and a small real set improved real-world performance by 38% on average over a policy trained on the real set alone [1]. Active system identification trains an exploration policy in simulation that maximizes the Fisher information of its real trajectory about the physical parameters [23]. Influence functions can rank real demonstrations by their effect on the closed-loop success of a policy and select the newly collected trajectories that help it most [24]. The work closest to mine uses a twin to find failure-prone configurations for real rollouts [14]. In these works the real data correct either the model [1],[24],[14] or the simulation [23], and all except [1] choose which real data to collect. To my knowledge, nobody has used the same selected real data to correct both the reconstruction of a visual digital twin and the model trained in it.

Finally, the generalization gap of visual manipulation policies trained by imitation can be split into factors of variation such as lighting or camera placement, and the ordering of these factors by difficulty was largely the same in simulation and on a real robot [25]. Active evaluation fits a probabilistic model over task factors such as object pose and camera viewpoint to choose informative real trials [26], but it predicts performance from real outcomes of the same policy and task, not from measurements taken before a method is applied. Few studies test whether properties of the shift measured before a method is applied, such as the change in internal features between twin and real views, predict whether its gain carries over to a new scene or to another task. It is also unclear whether such measurements do better than simple indicators such as the size of the gap. My research questions in §7 address these open points.

(Please see section 12 for citations used in this document)

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The aim of my dissertation is to develop representation learning methods that improve the generalization of deep learning models trained in digital twins built from real data. The models should work reliably in reality and in places not seen during training, and the methods should work with unchanged settings for both navigation and manipulation.



I will not have my own robot at the start and will train models on academic GPU clusters, so I will fine-tune existing pretrained models rather than train large policies from scratch. I will rely on public datasets that contain independent captures of the same real scenes. I will study two tasks, robot navigation and robotic manipulation, and measure all results on held-out scenes that were not used for training or model selection.

My research questions (RQ) and hypotheses (H) are as follows.

1. RQ1. Which reconstruction errors of a digital twin, varying across a scene, harm the generalization of navigation and manipulation models trained in it?
2. H1. The harm of an error in a region is the drop in success on held-out scenes of a model trained in a twin that contains this error, compared with the same model trained in a twin in which the error is corrected. This harm is predicted better by how much the error changes the features of a fixed pretrained visual encoder, on views rendered with and without it, than by its size in the image or geometry, as measured by PSNR, LPIPS or depth error. Also, randomizing each region according to its estimated reconstruction error and task relevance generalizes better than uniform randomization with the same average strength over the scene.
3. RQ2. At which stage of a model trained or fine-tuned in a twin do real inputs first lose task information that is available for simulated inputs? I expect linear probes checked against control tasks to identify an earliest stage or a narrow range of stages. I will test with causal interventions whether the model uses this information.
4. H2. Aligning simulated and real representations at that stage reduces the simulation-to-reality gap more than the same alignment at the input, at the final features, at all stages or at layers chosen by existing criteria, given the same real data and training effort.
5. RQ3. Which real observations, chosen under a fixed budget with the representation-based signals of RQ1 and RQ2, reduce the gap most when they are used both to refine the reconstruction of the twin and to fine-tune the model? I expect that correcting both with the same selected data reduces the gap more than correcting only one of them. I will compare my selection with random and failure-driven selection.
6. RQ4. Do the gains transfer with unchanged settings to unseen scenes and from one task to the other? Can the representation-based signals of RQ1 and RQ2, measured before a method is applied, predict this transfer better than the gap before adaptation or the image-level discrepancy between twin and real views, such as LPIPS?
7. Fallback. If the size of an error predicts harm better than the change in features, I will guide randomization and data selection with reconstruction uncertainty and action sensitivity instead. This fallback is weaker, because uncertainty mostly shows where the input views constrain the scene weakly and can stay low for an error that all views agree on.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / **Contribution of the expected results to development of scientific discipline**

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The main expected contribution of my dissertation is a set of representation learning methods that help deep learning models trained in digital twins generalize better to reality and to unseen places. I also want to learn whether most of the simulation-to-reality gap comes from a small number of reconstruction errors that matter for the task, whether their effect can be traced to a specific stage of the model, and whether a small amount of well-chosen real data is enough to reduce it.



The experiments should indicate which reconstruction errors harm generalization, at which stage of a model task information is lost on real inputs and when it pays off to correct both the twin and the model with the same real data. They should also show whether the gains carry over to new scenes and between navigation and manipulation. Because these questions concern learning under distribution shift, the results on where to align representations and which data to select may also be useful in domain adaptation and active learning. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods on held-out scenes in the same way. I hope the results will tell practitioners how accurate a digital twin must be for a given task and which real data are worth collecting, so that learning-based robots can be deployed in new places with less effort.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Most of my work will consist of building digital twins of real scenes, training and fine-tuning deep learning models in them, and designing methods that analyse and align the representations these models learn. No established theory covers the generalization of deep models trained in digital twins reconstructed from real data, so the work will be mostly empirical. Domain adaptation theory will be my motivation, and where possible I will relate my methods to the terms of the bound in [3], for example by checking whether an alignment at a given stage lowers the discrepancy term without raising the joint error term.

I will use PyTorch for model training, gsplat for 3D Gaussian Splatting [11], COLMAP for camera poses, the Habitat simulator for navigation and ManiSkill3 for manipulation. I will use Git with GitHub repositories for version control and Hugging Face for downloading and publishing models and datasets. I will start from pretrained models where possible. I will release the code and evaluation scripts, and also trained models and derived data where licences allow.

Since I will not have my own robot at the start, I will evaluate mainly on public data. For navigation I will use public datasets of real indoor scenes with independent captures of the same scenes, such as ScanNet++ and MuSHRoom. I will build the twin from one capture. The other capture gives real frames, paired with twin renderings from the same camera poses, for probing and alignment. Together with the laser scans it also gives a more accurate reference environment, in which I run the models closed-loop as a proxy for reality. This proxy mainly measures the part of the gap caused by reconstruction errors and leaves out differences in how a real robot moves and senses, such as its actuation and its own camera. For manipulation I will build twins of real tabletop scenes from images in public robot datasets, following the visual matching of SIMPLER [15]. I will then check whether the success rates of the same policies in these twins agree with their published real-robot success rates, both in value and in the ranking of the policies. If access allows, I will use a real robot in a university laboratory to check whether selected improvements also appear in reality and whether the reference environment ranks models in the same order as real runs.

For RQ1 I will change one type of reconstruction error at a time in selected regions. I will either correct it using a more accurate reference, such as the laser scans of ScanNet++, or inject it in a controlled way, and then measure the resulting change in the success rate of the trained model. I will compare the change in the features of a fixed pretrained encoder with PSNR, LPIPS and depth error by their rank correlation with harm across regions and scenes. I will also test region-wise randomization against uniform randomization with the same average strength over the scene (H1). I will estimate the error of a region from its feature change and its task relevance from how much the actions of the model change when the region is perturbed. For RQ2 I will train linear probes, that is, linear models that read task information such as the direction to the goal, on the hidden states of each stage for simulated inputs, and test them on the paired real inputs. Control tasks with random



labels will show that the probes do not simply memorize. To test whether the model uses the lost information, I will replace the component of the real hidden state along the probe directions with the one from the paired simulated input and check whether the action improves. I will compare this with replacing the same number of random directions. I will then add an alignment loss, such as maximum mean discrepancy or a contrastive loss between paired simulated and real features, at the identified stage. I will compare it with the same loss at the input, at the final features, at all stages and at layers chosen as in surgical fine-tuning [22] (H2). For RQ3 I will compare rules for selecting real data with random and failure-driven selection at equal budgets, and compare correcting both the twin and the model with correcting only one. For RQ4 I will apply the methods with unchanged settings to held-out scenes and to the other task. I will then check whether the signals of RQ1 and RQ2, measured before adaptation, predict the gains, using Spearman rank correlation across scenes with permutation tests.

On held-out scenes I will measure the success rate of each model in the twin and in the evaluation environment of each task. For navigation this is the reference environment, and for manipulation it is the published real-robot results and, if access allows, a real robot. The simulation-to-reality gap is the difference between the success rates in the twin and in this environment. Following [2] and [15], I will also measure how well results in the twin predict the ranking of models in the evaluation environment. I will compare my methods with domain randomization, domain-adversarial training, sim-and-real co-training and the baselines named above, using the same number of training steps and the same amount of real data for every method. To get reliable results I will repeat runs over random seeds and scenes, use statistical tests for comparisons and run ablation studies. Data for training, model selection and final evaluation will be kept separate.

I will run experiments on the supercomputers of the Wrocław Centre for Networking and Supercomputing (WCSS), with jobs managed by SLURM, and on PLGrid resources. I intend to use AI coding assistants like Claude Code to speed up writing and testing code. They will also help with routine work such as monitoring long experiments and proofreading.

I plan to publish my results at conferences and in journals worth 200 points on the ministerial list. In machine learning these are NeurIPS, ICML and ICLR, in computer vision CVPR, ICCV and ECCV, and in robot learning RSS and the journal IEEE RA-L. Where possible, I will also post the papers as arXiv preprints.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / *The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

Streszczenie popularnonaukowe

Roboty coraz częściej uczą się w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań, ponieważ nauka w prawdziwym świecie jest powolna, kosztowna i czasem niebezpieczna. Taka kopia nigdy nie jest idealna. Na przykład szklane drzwi mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. Chcę ustalić, które różnice między kopią a rzeczywistością szkodzą modelowi i w którym miejscu wewnątrz modelu zamieniają się w błędy. Chcę też sprawdzić, które nieliczne rzeczywiste przykłady najlepiej poprawiają zarówno model, jak i kopię. Zbadam to w nawigacji robotów oraz w chwytaniu i przestawianiu



przedmiotów. Efektem mają być metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach.

Abstract for general public

Robots are increasingly trained in digital twins, that is, virtual copies of real places rebuilt from their recordings, because learning in the real world is slow, costly and sometimes unsafe. Such a copy is never perfect. For example, a glass door may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. I want to find out which differences between the copy and reality harm the model and where inside the model they turn into errors. I also want to check which few real examples best correct both the model and the copy. I will study this in robot navigation and in grasping and moving objects. The result should be methods that let robots learn to work in new places faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

September 2027 (planned submission to a venue worth 200 points on the ministerial list, such as the ICLR 2028 conference or the journal IEEE Robotics and Automation Letters).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

I used Claude (via Claude Code) for literature search, drafting and editing, and I reviewed all content and references myself.

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13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

My supervisor and I plan to meet weekly. At these meetings I will present my progress and recent results, and we will agree on the next steps and discuss our joint papers. I will also attend the weekly meetings of my supervisor's group of PhD students, where we discuss current results, problems and new work in the field. For daily contact we will use email and instant messaging. Code, trained models and the LaTeX sources of papers will be kept in shared Git repositories.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor



Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor